

EMC ASSESSMENT (EU – RADIO EQUIPMENT DIRECTIVE)

Product: SidecarTridge Croissant Keyboard Emulator

Model/Revision: SidecarTridge Croissant Keyboard Emulator for Atari ST/STe / REV1.1.0

1. PRODUCT OVERVIEW

The product is a low-voltage digital electronic device intended for use in retro computing systems. It is designed as an internal or semi-internal component (e.g. expansion board, cartridge, or interface module) and is powered by the host system.

The product integrates a radio module providing wireless functionality (e.g. Bluetooth and/or Wi-Fi) in the 2.4 GHz band.

Typical operating voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

2. APPLICABLE DIRECTIVE

This assessment is made in relation to:

- Directive 2014/53/EU (Radio Equipment Directive – RED)

The essential requirements of:

- Article 3.1(a) – Health and safety
- Article 3.1(b) – Electromagnetic compatibility
- Article 3.2 – Efficient use of radio spectrum

are applicable to this product.

Directive 2014/30/EU (EMC) and Directive 2014/35/EU (Low Voltage) do not apply in addition to Directive 2014/53/EU.

3. DESIGN CHARACTERISTICS RELEVANT TO EMC AND RADIO PERFORMANCE

- The device contains standard digital logic components (e.g. CMOS/TTL, microcontrollers, memory devices).
- The product integrates a radio module operating in the 2.4 GHz ISM band.
- The radio functionality (e.g. Bluetooth and/or Wi-Fi) is provided entirely by the integrated module.
- No modifications affecting radio frequency performance (antenna, RF path, shielding, or firmware parameters) have been made to the module.

- The device does not incorporate high-frequency switching power supplies (unless provided externally by the host system).
 - Signal frequencies of non-radio circuitry are limited to those typical of embedded or legacy computing systems.
 - External connections are limited in length and are typically internal to the host system or via short cables.
 - The device includes a microcontroller-based system operating at moderate clock frequencies typical of embedded systems.
 - A USB interface is present for local data transfer; it is not intended for continuous high-speed communication during normal operation.
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4. RADIO MODULE INTEGRATION AND EMC ASSESSMENT

The product integrates a pre-assessed radio module providing wireless communication functionality.

Compliance with the essential requirements of Directive 2014/53/EU is ensured through the use of this module in accordance with the module manufacturer's integration guidelines.

The EMC performance of the radio functionality is supported by harmonised standards such as:

- EN 301 489-1 (EMC for radio equipment – general)
- EN 301 489-17 (EMC for broadband data transmission systems)

The host product:

- does not introduce additional continuous high-frequency emission sources
- does not alter the RF characteristics of the integrated module
- operates as a low-power internal digital system
- The USB interface is used intermittently for file transfer and does not represent a continuous high-frequency emission source.

Conclusion:

The product does not adversely affect the EMC performance of the integrated radio module when used as intended.

5. RF EXPOSURE

Compliance with RF exposure requirements is ensured through:

- EN 62479 (assessment of low-power equipment)

The product operates at power levels consistent with the integrated radio module and is intended for use within enclosed host systems.

Conclusion:

The product meets the applicable requirements for human exposure to electromagnetic fields.

6. INSTALLATION AND USE CONDITIONS

- The product is intended for integration into a larger electronic system.
 - It is not designed to operate as a standalone end-user device.
 - Proper EMC and radio performance depend on correct installation and use in accordance with the module manufacturer's integration guidelines.
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7. OVERALL CONCLUSION

Based on the use of a pre-assessed radio module and the design characteristics of the product, the device is considered to comply with the essential requirements of Directive 2014/53/EU when installed and used as intended.

This assessment is supported by harmonised standards such as:

- EN 300 328 (radio spectrum use)
- EN 301 489-1 and EN 301 489-17 (EMC for radio equipment)
- EN 62479 (RF exposure)

No additional EMC mitigation measures are required based on the design and intended use.

8. RESPONSIBLE PERSON

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Date: 23 April 2026

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